

P4频谱探测器

P4 Spectrum Detector

规格书

Specification Sheet

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1 功能用途

1 Function and Purpose

UAMS-D02 无人机侦测定位探测器可通过解析通信链路实现对无人机探测和飞手的定位，具有黑白名单和轨迹追踪的功能，可灵活部署，适用于多场景和环境。

UAMS-D02 drone detection and positioning detector can achieve drone detection and pilot positioning by analyzing the communication link. It features a blacklist and whitelist function as well as trajectory tracking, allowing for flexible deployment and is suitable for various scenarios and environments.



1.1 产品特点

1.1 Product Features

1) 飞手定位、轨迹追踪

1) Drone pilot positioning、Trajectory Tracking

可以展示无人机实时飞行轨迹、遥控频段、型号、唯一 ID、飞行高度速度、无人机位置、飞手位置等信息。

D02 can display real-time flight trajectories of drones, remote control frequency bands, models, unique IDs, flight altitudes and speeds, locations of drones, and locations of pilots.

2) 与多雷达、诱骗等设备联动

2) conjunction with radars, GPS deception

提供扩展接口，可方便实现与雷达、GPS 诱导、干扰等设备进行联动。

D02 provide extended interfaces, which can conveniently realize the linkage with radar, GPS guidance, interference and other devices.

3) 黑白名单

3) Blacklist and Whitelist

通过预定义或动态学习无人机信号特征，实现合法设备放行与非法目标拦截的精准管控，有效降低误报率，提升安全防护效率。

By pre-defining or dynamically learning the signal characteristics of drones, precise control and management can be achieved to allow passage for legitimate devices and to intercept illegal targets, effectively reducing the false alarm rate and enhancing the efficiency of security protection.

4) 灵活部署、操作简单

4) Flexible deployment and Ease of Operation

频谱重量轻，可单人应用架设，方便用户使用；同时人机界面直观简洁，配

置简单。

D02 is lightweight, can be set up by a single person, making it convenient for users. Additionally, the human-machine interface is intuitive and straightforward, and the configuration is simple.

1.2 产品指标

1.2 Product Specifications

1) 频率范围：30MHz—6000MHz;

1) Frequency range: 30MHz—6000MHz;

2) 探测距离：5-10km（无人机发射功率 0.1W标准场地）；

2) Detection range: 5-10km (standard field with a drone transmission power of 0.1W);

3) 扫描速度：20GHz/s (25kHz 分辨率)

3) Scan speed: 20GHz/s (25kHz resolution);

4) 识别技术：频谱特征、定位解析(包括 O4 解析)、RID 解析；

4) Identification technology: spectral features, positioning analysis (including O4 analysis), RID analysis;

5) 极化方式：垂直极化；

5) Polarization method: Vertical polarization;

6) 测向精度： $\leq 5^\circ$ (RMS 标准场地)；

6) Range accuracy: $\leq 5\text{m}$ (RMS standard site);

7) 定位精度： ≤ 10 米；

7) Positioning accuracy: $\leq 10\text{ m}$;

8) 同时探测目标数： ≥ 30 架（同时定位飞手数量 ≥ 10 个）；

8) Simultaneous target detection: ≥ 30 aircraft (simultaneous positioning of pilots ≥ 10);

9) 设备功率: $\leq 100\text{W}$;

9) Equipment power consumption: $\leq 100\text{W}$;

10) 结构尺寸 $450\text{mm} * 450\text{mm} * 380\text{mm}$;

10) Structural dimensions $450\text{mm} * 450\text{mm} * 380\text{mm}$;

11) 重量: $\leq 13\text{Kg}$;

11) Weight : $\leq 13\text{Kg}$;

12) 供电方式: 220V AC 交流电;

12) Power supply method: 220V AC alternating current;

13) 数据接口: 网口;

13) Data Interface: Network Port;

14) 防护等级: IP65;

14) Protection level: IP65;

2 场景测试

2. Scenarios Test

